

Feiyang Chen

BACKEND & AI APPLICATION ENGINEER

Suzhou, Jiangsu, China

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Summary

Backend and AI application engineer with three years of Linux kernel and systems experience and end-to-end AI application delivery. Independently delivered a large language model (LLM) news proofreading agent and owned the full-stack administration system for Suzhou News. Experienced in Python/FastAPI, LangGraph, retrieval-augmented generation (RAG), Model Context Protocol (MCP), asynchronous workflows, data pipelines, containerised deployment, and observability.

Skills

Languages	Python, Java, C/C++, TypeScript
AI & Backend	FastAPI/REST API, SQLAlchemy, LLM, LangChain/LangGraph, RAG, MCP, DeepEval
Data & Tasks	PostgreSQL/pgvector, Redis, PgQueueer, MinIO
Engineering	Docker, Kubernetes, GitHub Actions, OpenTelemetry, Prometheus/Grafana
Systems	Linux Kernel, MIPS, LoongArch

Projects

Suzhou Xinsu Newspaper Network Technology Co., Ltd.

Suzhou, Jiangsu, China

SOFTWARE ENGINEER (PROJECT COLLABORATION)

Apr. 2026 – Aug. 2026

- **News Proofreading Agent (Apr.–May, independent):** Owned requirements, architecture, and end-to-end delivery; built a Vue editor SDK and FastAPI/SQLAlchemy/LangChain backend for agent orchestration, fact retrieval, and revision generation.
- Deployed Qwen locally, designed asynchronous execution and error handling, and used Langfuse to trace model calls, task states, and end-to-end workflows.
- **Industrial Drawing/Circuit-Diagram Recognition:** Reproduced research methods and contributed to data annotation, training, and evaluation of YOLO, UNet, and HRNet models.
- **Suzhou News Administration System (Jun.–Aug., two-person team):** Owned the full-stack RuoYi-Vue-Plus (Spring Boot/Vue 3) administration system and integration with the public portal.
- Implemented multi-tenant content management, video compression, legacy data migration, third-party synchronization, and scheduled static-site builds.

A-Share Fundamental Analysis Agent

Suzhou, Jiangsu, China

SOFTWARE ENGINEER (PERSONAL OPEN-SOURCE PROJECT)

Feb. 2026 – Mar. 2026

- Built an A-share scoring and Q&A system with FastAPI, SQLAlchemy, and PgQueueer, including CNInfo/Yahoo Finance ingestion, asynchronous scheduling, and declarative YAML scoring rules.
- Implemented a streaming LangGraph + RAG + SSE pipeline with MCP, PostgreSQL/pgvector, and MinIO; added DeepEval offline regression suites for RAG, agent, and end-to-end workflows.
- Validated dependencies with pytest/Testcontainers; provided Docker Compose and Kustomize/Minikube deployment configurations and integrated OTel, Prometheus, Grafana, Loki, and Tempo.

Systems Software and Open-source Contributions

Suzhou, Jiangsu, China

KERNEL ENGINEER

Jun. 2020 – Aug. 2023

- Added LoongArch64/Linux support to Valgrind; completed core porting, instruction translation, and runtime validation, reaching initial usability in one month and passing hundreds of regression tests.
- Adapted Linux, klibc, Kcov, and RT-Thread for LoongArch and supported LS2K0300 integration.
- Enabled OpenStack Nova/DevStack on Fedora MIPS and brought up the core service workflow within one month.
- Designed documentation CI/CD with live PR previews and automatic contributor attribution.

Work Experience

Loongson Technology Corporation Limited

Suzhou, Jiangsu, China

KERNEL ENGINEER

Dec. 2020 – Aug. 2023

- Submitted upstream MIPS/LoongArch Linux patches, covering root-cause analysis, fixes, builds, and regression testing.
- Led LoongArch enablement for klibc, Valgrind, and other low-level software; resolved build, instruction, runtime, and performance issues.
- Improved GitHub build, test, and release workflows, increasing delivery reliability and team efficiency.

Jiangsu Lemote Technology Co., Ltd.

Suzhou, Jiangsu, China

KERNEL ENGINEER INTERN

Jun. 2020 – Dec. 2020

- Developed and debugged Linux kernel code through issue reproduction, log analysis, patching, and validation.
- Ported OpenStack Nova and DevStack to Fedora MIPS, resolving compatibility blockers and validating deployment.

Education

The University of Sydney

Sydney, NSW, Australia

MASTER OF COMPUTER SCIENCE

Feb. 2024 – May 2026

- **ISSRE 2026 accepted paper (second author):** “Dr. DD: 1-Minimal Isolation of Failure Causes via Deferred Restarts.”

Suzhou University of Technology (formerly Changshu Institute of Technology)

Suzhou, Jiangsu, China

BACHELOR OF COMPUTER SCIENCE AND TECHNOLOGY

Sep. 2016 – Jun. 2020

- **ICAIS 2020 paper (first author):** “Formal Verification of the Correctness of Operating System on Assembly Layer in Isabelle/HOL.”